

Addendum: Permutation Calibration of the Bounded Asymmetry Profile, Withdrawal of One Localisation Claim, and a Reporting Requirement for Unsigned Density Contrasts

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July 28, 2026

1. Scope and status of this document

This addendum accompanies the record `10.5281/zenodo.21627908` and the code repository at

<https://github.com/innerlightr-wq/disprot-bounded-asymmetry>

It documents the substantive changes between the original report on bounded asymmetry profiles in DisProt-curated protein disorder data and the revised manuscript titled *A permutation-calibrated bounded asymmetry profile, and a negative cross-taxon concordance result for an AlphaFold-derived disorder proxy*.

Three points about its status:

1. It is a record of changes, not a replacement for either version. The revised manuscript is self-contained and should be read in preference to this document for any substantive question.
2. It contains no figures. Two figures belonging to the revised manuscript — a panel of asymmetry profiles overlaid with permutation null bands, and a null-calibration figure for the unsigned magnitude index — are still pending final regeneration. Every number they would display is available in tabular form in the revised manuscript and in `permutation_results.csv`; nothing in this addendum depends on them.
3. One claim made in the original version is **withdrawn**, and it is the claim the original version treated as its methodological headline. That withdrawal is set out in Section 6 and is the main reason this addendum exists.

2. What prompted the revision

The original version separated its descriptive apparatus from its inferential apparatus in principle but not in practice. Three scalar summaries of the profile — the unsigned magnitude $\overline{A}$, the signed balance $\overline{B}$ and the directional consistency $\mathcal{C}$ — were reported as data descriptions, but the surrounding text read a particular pattern among them as evidence that two distributions differed in a way global two-sample tests had missed. No null distribution had been computed for any of the three.

An independent statistical audit supplied that null distribution. The result is that $\overline{A}$ carries a large, design-dependent positive bias under the null hypothesis, that the value on which the original localisation claim rested falls inside its own null distribution, and that the profile statistics were in this dataset strictly less sensitive than the classical tests they were offered as a complement to.

3. Inferential procedures added

Four procedures were added to the analysis, and their design decisions matter enough to state here.

Permutation test. The null $H_0 : F_A = F_B$ is tested by reshuffling the pooled sample into groups of the original sizes, with 2,000 replicates per comparison and $p = (b+1)/(B+1)$, giving an attainable minimum of $p = 5 \times 10^{-4}$. Two decisions are load-bearing. First, the *entire* estimation pipeline, bandwidth selection included, is recomputed inside every replicate. Fixing a bandwidth from the observed data would break the correspondence between the null distribution and the reported estimator, and would conceal rather than calibrate the differential-smoothing behaviour that turns out to drive the bias in $\bar{A}$. Second, group sizes are preserved, so the unequal- n bandwidth mismatch is present under H_0 exactly as it is in the data. Statistics tested are $\bar{A}$, the density-weighted magnitude A_w , and the trimmed supremum $T_{\max}$.

Mass-trim region. Supremum statistics and band interpretation are restricted to the region retaining 99% of pooled probability mass. Outside it both densities are numerically negligible and the profile is a ratio of two estimator artefacts. No claim is made there.

Bootstrap simultaneous band. A standardised percentile-of-supremum band, with a floor on the point-wise scale to prevent grid points of collapsing variance from dominating the supremum. The band covers S_h , the profile the selected bandwidth would produce from infinite data — not the underlying asymmetry. Smoothing bias is unrepresented, and near the domain boundaries the renormalised Gaussian estimator is biased, so coverage there is not nominal.

Permutation null band. A familywise-controlled local significance band over the retained-mass region. Where the observed profile leaves this band, the local departure is significant at level α simultaneously over that region. This is the only construct in the study that licenses a statement about *where* a difference lies.

Permutation p -values are Holm-adjusted within each family of six pairwise comparisons for a given feature and statistic. Holm is valid under arbitrary dependence, which matters because the six comparisons share the same four samples.

4. Definitional changes

A density-weighted magnitude. The revised manuscript adds

$$A_w = \frac{\int_{\Omega} |S(x)| \bar{p}(x) dx}{\int_{\Omega} \bar{p}(x) dx}, \quad \bar{p} = \frac{1}{2}(p_A + p_B),$$

which weights each point by the mass actually present there rather than equally. As $\varepsilon \rightarrow 0$ it converges exactly to the total-variation distance between the two density estimates. A_w is therefore not a new quantity but a familiar one recovered in the profile's coordinates, which is why it behaves better.

Reparameterisation equivariance. The regularisation constant ε carries the units of a density, so the profile as originally written was not invariant under rescaling of the feature coordinate. Setting $\varepsilon = \delta/|\Omega|$ with δ dimensionless restores equivariance. On the unit domain used throughout this study the substitution returns δ unchanged, so **no reported numerical value is affected**; only the definition becomes well posed.

Explicit descriptive/inferential separation. The profile itself, its zero crossings, the three scalar summaries and A_w are declared descriptive: functions of the data with no attached error statement, none of which supports a claim that two populations differ. The permutation test, the two bands and the classical tests are declared inferential. The revised manuscript adheres to this separation throughout, and the language sweep that enforced it removed every occurrence of *reveals*, *demonstrates* and *missed by*.

5. Disposition of the original claims

Every substantive claim of the original version is listed below with its disposition under the audit. Two are withdrawn, one is withdrawn as an inferential statement while surviving as a description, one is rewritten, one is reversed, and the rest stand.

Table 1: Claim-by-claim disposition. Permutation p -values are Holm-adjusted within families of six unless marked otherwise. “Table 2” in the Basis column refers to the revised manuscript; the section cross-reference is internal to this addendum.

Claim in the original version	Status	Basis
The profile resolves structure that global tests miss (section heading and principal methodological claim)	Withdrawn	Its sole support was the Human- <i>E. coli</i> comparison, which does not reject; see Section 6
Human- <i>E. coli</i> on curated disorder shows localised structure ($\bar{A} = 0.143$, $\mathcal{C} = 0.57$)	Withdrawn	Observed $\bar{A}$ lies inside its own permutation null (median 0.111, 95th percentile 0.174), raw $p = 0.18$
Two zero crossings indicate regions of opposing local enrichment that cancel in aggregate	Withdrawn as inferential	No band excludes zero anywhere in this comparison; the count is retained in Table 2 as descriptive only, flagged as such in the caption
A null global test does not imply locally similar densities	Rewritten	What the data establish is that the two <i>fitted</i> densities differ; the general possibility remains real but is not demonstrated here
Fungal proteins enriched in curated disorder relative to human	Retained	Now supported under permutation: A_w Holm $p = 3 \times 10^{-3}$; KS $p = 7 \times 10^{-4}$
Fungal proteins enriched relative to bacterial	Retained	$\bar{A}$ Holm $p = 3 \times 10^{-3}$, A_w Holm $p = 1.0 \times 10^{-2}$; KS $p = 3 \times 10^{-3}$
The AlphaFold very-low fraction is weakly concordant with curated disorder and taxon-dependent in its off-set	Retained, promoted to lead result	Unchanged numerically; all three bacterial comparisons on this feature are significant on every statistic tested, Holm $p = 3 \times 10^{-3}$
The profile adds sensitivity beyond scalar two-sample testing	Reversed	The set of comparisons it rejects is a strict subset of the set rejected by Kolmogorov-Smirnov

6. The withdrawn claim, in detail

The original version presented the human versus *E. coli* comparison on curated disorder fraction as its worked demonstration: global tests are null (KS $p = 0.39$; Mann-Whitney $p = 0.85$; rank-biserial -0.009), yet the profile returned $\bar{A} = 0.143$ with $\mathcal{C} = 0.57$ and two zero crossings, which was read as localised structure that global testing had missed.

Permutation calibration does not support that reading. Applied to these group sizes — 1,339 human against 146 bacterial proteins — this pipeline produces, under the null, $\bar{A}$ with median 0.111 and 95th percentile

0.174. The observed 0.143 falls inside that distribution:

Statistic	Observed	Null median	Raw p
$\bar{A}$	0.143	0.111	0.18
A_w	0.086	0.079	0.40
$T_{\max}$	0.350	—	0.36

All three Holm-adjusted values are ≥ 0.53 . The bootstrap interval for the signed index likewise includes zero ($\bar{B} = 0.081$, 95% CI $[-0.036, 0.250]$). The profile structure in this comparison is what the estimation pipeline generates from two samples of these sizes drawn from a single distribution. No departure from the null is demonstrated, and the figure that displayed this comparison should not be read as showing one.

A separate point, unaffected by the audit, is worth restating: a null human–*E. coli* contrast in curated disorder is a product of DisProt ascertainment rather than evidence against the prokaryote–eukaryote disorder gap reported from proteome-wide prediction. *E. coli* entries exist in DisProt because their disorder was investigated, which selects against the ordered majority of that proteome.

7. The replacement result: null calibration of the scalar indices

The audit produced a finding the original version did not contain, and it is more transferable than anything it displaced, because it is a property of the construction rather than of these data.

$\bar{A}$ is substantially biased away from zero under the null, by an amount that depends on the design.

Across the twelve comparisons its null median ranges from 0.089 to 0.398. It is largest where a small group is compared with a larger one and where the underlying densities are peaked: for the bacterial–nematode comparison on the AlphaFold feature (146 against 58 proteins), samples drawn from one distribution routinely produce $\bar{A} \approx 0.4$. In four of the twelve comparisons the observed value falls *below* its own null median.

The mechanism is differential smoothing. Scott’s rule selects a wider bandwidth for the smaller sample, so the two density estimates differ systematically in resolution even when they estimate the same density. The profile is then bounded away from zero over much of the domain, and $\bar{A}$, which integrates $|S|$ with equal weight everywhere including regions of negligible mass, accumulates that difference. An observed $\bar{A}$ therefore carries no interpretation on its own, is not comparable between comparisons with different group sizes, and is not comparable between features whose densities have different shapes.

$\bar{B}$ does not share this behaviour. The smoothing discrepancy perturbs the profile pointwise with sign that varies over the domain, so it cancels in the signed integral while accumulating in the unsigned one. Empirically the bootstrap intervals for $\bar{B}$ and the permutation verdicts agree in all twelve comparisons: the five intervals excluding zero are exactly the five comparisons that reject, and the seven including zero are exactly the seven that do not. This is an empirical agreement on twelve non-independent comparisons, not a proof of unbiasedness; establishing it generally would need a simulation study over a grid of group sizes.

A_w is intermediate and preferable to $\bar{A}$. Its null medians span 0.051–0.117, a range less than one third as wide, because weighting by the pooled density suppresses precisely the low-mass regions where the smoothing artefact lives. It is also the more sensitive statistic in three of the four rejected families. Given its total-variation limit, this amounts to the observation that total-variation distance is the better-behaved magnitude, which is not surprising.

$\mathcal{C}$ inherits the bias in its denominator and is therefore attenuated in exactly the small-sample comparisons where a reader would most want a stability diagnostic. It is reported for completeness and no conclusion

is drawn from it.

The resulting reporting requirement

Any use of an unsigned normalised density contrast of the form $(p_A - p_B)/(p_A + p_B)$ should accompany the observed magnitude with its permutation null for the same group sizes and the same estimation pipeline. Reporting the bandwidth rule alone — which is what the framework was originally specified to require — is not sufficient. Every result in the revised manuscript’s permutation table was produced under a fixed, reported bandwidth rule, and the bias is present anyway.

This is the recommendation with the widest reach, since normalised density contrasts of this form appear in several applied literatures. The failure is not exotic: it appeared here at conventional sample sizes with a standard bandwidth rule, and it was invisible until the pipeline was permuted.

8. Power relative to classical tests

The revised manuscript reports, and this addendum records, that on these data the profile statistics were *less* sensitive than the classical tests. Two comparisons are rejected by Kolmogorov–Smirnov and not by the profile after adjustment: fungal versus nematode on curated disorder (KS $p = 6 \times 10^{-3}$; A_w Holm $p = 0.138$) and human versus fungal on the AlphaFold feature (KS $p = 0.018$; A_w Holm $p = 0.71$). Across the twelve comparisons the set rejected by the profile is a strict subset of the set rejected by KS.

The profile is accordingly described in the revised manuscript as a device for localising and summarising a difference already established by a global test, not as a means of detecting one. Its one genuine contribution in this study was to attribute the bacterial departure on the AlphaFold feature to the mass at and near zero rather than to the distribution as a whole — a familywise-controlled local statement, available because the trimmed supremum was significant in those three comparisons and in no others.

9. What did not change

- All descriptive numerical values. The indices, bootstrap intervals, classical test results, concordance statistics and robustness values are as originally computed, from the same four DisProt release 2026_06 per-organism JSON exports covering 1,766 proteins.
- The principal substantive finding. The AlphaFold-derived very-low-confidence fraction remains weakly concordant with curated disorder within every taxon examined (Spearman $\rho = 0.07$ – 0.24) and taxon-dependent in its offset (median signed gap $+0.001$ human against -0.083 *E. coli*), and should not be used as a taxon-neutral protein-level disorder feature without recalibration.
- The reconciliation with the pLDDT literature. The very-low band is defined at pLDDT < 50 against an empirically optimal disorder cutoff near 68.8, so this is read as a property of a particular precomputed field rather than as a contradiction of established residue-level performance.
- Every limitation previously stated: ascertainment bias, the absence of residue-level evidence, the point masses at zero, the negative-residue assumption, and the unresolved provenance items arising from blocked database access.

10. Limitations of the audit itself

- 2,000 permutations give a p -value floor of 5×10^{-4} , adequate for the families used but not for larger ones. The permutation results should be treated as reproducible diagnostics to be re-run at higher B if a finer resolution is needed.
- Holm was applied within families of six. Under a single family of thirty-six, the three bacterial comparisons on the AlphaFold feature remain significant on all statistics, fungal–bacterial on curated disorder remains significant on $\bar{A}$ and A_w , and human–fungal remains significant on A_w ; but human–fungal on $\bar{A}$ and fungal–bacterial on $T_{\max}$ would not. No conclusion drawn depends on those two.
- The good behaviour of $\bar{B}$ is an empirical observation on twelve non-independent comparisons, not a theoretical result.
- The simultaneous bands cover the bandwidth-smoothed estimand, not the underlying asymmetry, and coverage is not nominal near the domain boundaries.
- Both features have atoms at zero, and kernel density estimation of a distribution with atoms is misspecified. The permutation null is computed under the same misspecification, so the test remains valid, but the bands near zero should not be read as estimates of a smooth object.

11. Materials accompanying this addendum

Added to the record alongside the revised manuscript:

- `profile_inference.py` — permutation test, bootstrap simultaneous band, permutation null band, mass-trim mask, the A_w and $T_{\max}$ statistics, and Holm adjustment. Validity-critical choices are marked in the source. The two that must not be altered are the recomputation of the full pipeline inside every replicate and the preservation of group sizes under permutation.
- `permutation_results.csv` — observed statistics, null medians and upper percentiles, and raw and Holm-adjusted p -values for all twelve comparisons.

Still pending, and the reason a complete manuscript version has not yet been released: regeneration of the asymmetry-profile panel with permutation null bands overlaid, and generation of the null-calibration figure. Both are presentational. The quantities they will display are already reported in Tables 2 and 3 of the revised manuscript and in `permutation_results.csv`, and no conclusion awaits them.

How to cite this addendum: De Jesús, E. (2026). *Addendum: permutation calibration of the bounded asymmetry profile, withdrawal of one localisation claim, and a reporting requirement for unsigned density contrasts*. Deposited with 10.5281/zenodo.21627908.

Acknowledgement. The permutation inference reported here originated in an independent statistical audit of the earlier draft conducted with large language model tools in a cross-checking configuration. All numerical values were computed by the accompanying scripts from the deposited input files.